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Modelling green multimodal transport route performance with witness simulation software

Qing-Zhou Wang [a], Jia-Meng Chen [a], Ming-Lang Tseng [b, c, *], Hai-Min Luan [a], Mohd Helmi Ali [d]

[a] School of Civil and Transportation Engineering, Hebei University of Technology, Tianjin, 300401, China
[b] Institute of Innovation and Circular Economy, Asia University, Taichung, 41354, Taiwan
[c] Department of Medical Research, China Medical University Hospital, China Medical University, Taiwan
[d] National University of Malaysia Faculty of Economic and Management, Malaysia

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ABSTRACT

This study proposes an approach for dynamically calculating multimodal transport line parameters based on a "Witness" software simulation. The approach is designed to reduce the carbon emissions of multimodal transport, as green transportation has recently received increased attention. A multi-objective multimodal transport efficiency evaluation framework is combined with an order relation analysis and an entropy-weight fuzzy analytical hierarchy process method to assess route performance under various multimodal transport schemes. This study validated the proposed approach using the China-Europe railway network from Wuhan, China to Berlin, Germany. The results are as follows: (1) the energy conservation and emission reduction values of a multimodal transport scheme fluctuate significantly, and it is scientifically appropriate to increase the energy conservation and emission reduction values as the indexes for scheme planning; (2) a multi-objective and multi-scenario efficiency evaluation method can provide a solution for green multi-modal transport route planning, and can reduce energy usage and emissions by 24% as compared to an optimal routing considering a single objective; and 3) the China-Europe railway express has unique advantages in a multi-modal transport network, as it can save more than 60% of the transport time and reduce the carbon emissions by 40% as compared with a traditional water transport. This study presents a green multi-modal transport organization approach.
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1. Introduction

Global carbon dioxide emissions have reached 33.1 GT, and the transportation industry contributed over 30% of the carbon emissions in 2018 (Global Energy & CO_2 Status Report. 2018), becoming one of the main sources of carbon emissions. With the acceleration of globalization and the development of international trade, there is massive demand for transnational transport. The low-carbon/ "green" development of transportation has attracted increased attention from the government and industry. This study combines the advantages of various transport modes for higher economic benefits and reduced carbon emissions, as compared with single modes of transport and multimodal transport (Zhang et al., 2017). This study presents a major transport mode for long-distance and transnational transport, and promotes carbon reduction and energy saving in the international transportation industry, through incorporating the development concept of "green transportation" into the planning of multimodal transport. (Wang et al., 2019a, b).

Efficiency evaluation and route selection are the keys to multimodal transport network planning, and involve selecting different evaluation indicators for model construction and optimization. The current literature mainly focuses on the economic benefits of transport, and the transport route is selected by measuring the transport time and cost. Yang et al. (2011) proposed a cost model and a goal programming model for investigating China-India freight routes, to select advantageous routes based on transport cost and time. Resat and Turkay (2015) constructed a dual-objective mixed integer programming model for optimizing multimodal transport lines with the lowest transport time and cost. Increased

* Corresponding author. Institute of Innovation and Circular Economy, Asia University, Taichung, 41354, Taiwan.
E-mail addresses: 2006126@hebut.edu.cn (Q.-Z. Wang), 201731604013@stu.hebut.edu.cn (J.-M. Chen), tsengminglang@gmail.com, tsengminglang@asia.edu.tw (M.-L. Tseng), 201731604023@stu.hebut.edu.cn (H.-M. Luan), mohdhelmiali@ukm.edu.my (M.H. Ali).

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attention has been paid to the efficiency of energy saving and emission reductions in transport lines, concurrently with the development of efficiency evaluations for multimodal transport organization (Lv et al., 2019). Conventional multimodal transport network planning is only concerned with the optimal time and transportation cost index, and therefore has been unable to meet the development needs of green transportation. Relevant scholars have begun to use carbon emissions as an evaluation index for road network planning, to promote energy saving and emission reduction in transport routes. Sun et al. (2015) established a node-arc-based mixed integer nonlinear programming model for solving a multimodal transport line planning problem by using time windows as constraints, and converting carbon emissions into transport costs. Lam et al. (2016) discussed carbon emissions as a constraint, and established a dual-objective optimization model to optimize transport routes. The aforementioned studies mainly treat carbon emissions as part of the transport cost or as a constraint for the optimization model, rather than an independent evaluation index. These approaches cannot illustrate an evaluation index for green multimodal transport, as they did not pay enough attention to such an evaluation index. Multimodal transport involves influencing attributes, and is a complex transport process. A single-objective optimization model has difficulty meeting the demands of an actual transport scenario. It is necessary to generate a combined green evaluation index to establish a multi-objective and multi-scenario evaluation model for multimodal transport that involves the preferences of multimodal transport participants. This is particularly significant, as reasonable route planning is very important for low-carbon transportation systems, even more so than the formulation of relevant policies and regulations (Benjaafar et al., 2013).

Prior studies used heuristic algorithms for the simulation and optimization of multimodal transport route selection. Cho et al. (2012) discussed transport cost and time, and proposed a dynamic programming algorithm for investigating the multimodal transport route from Busan to Rotterdam. Zhang et al. (2013) explored the structure of a Dutch container transport network and environmental costs using genetic algorithm optimization. To improve the solution accuracy, some scholars have used hybrid algorithms. Zhang et al. (2016) proposed a mixed-integer programming model, and Wan (2019) used a hybrid algorithm comprised of a genetic algorithm and an ant colony algorithm to reduce the number of iterations and improve the operational efficiency of multimodal transport route selection. Studies have focused on the optimization algorithm design process or the hybrid algorithm to improve the speed and accuracy of the algorithm. In conventional solutions, the basic transport parameters of each transport node are usually set in a general way, which ignores the uniqueness and differences in the parameters of the same transport route in various transport schemes. The resultant errors reduce the accuracy of the choice for the optimal routes. By optimizing the model and improving the solution algorithm, the errors caused by dynamic changes of parameters in transport schemes can not be effectively reduced. Witness simulation software is widely used in management and optimization for production and logistics systems, simulations and planning of enterprise logistics, as well as for modelling and optimization of a supply chain, and can effectively simulate the processes of factories and workflows. Galankashi et al. (2016) developed a comprehensive simulation and model with the Witness simulation software, which was used to optimize a gas station queuing system and improve the sales rate. Wilson et al. (2016) employed "Microsoft Visual Basic" to control the Witness software to dynamically simulate the production process of a car engine, to thereby predict energy usage and potential energy savings opportunities. As the Witness simulation can dynamically adjust parameter changes, the influence of real-time parameter changes on the simulation results improves the simulation accuracy. Witness is also superior in realizing the simulation of parameter changes in the transportation process and in accurately simulating a multimodal operation process, and is able to reduce the errors in stimulation caused by the setting of general parameters.

The "Multimodal Transport Road Network" (MTRN) is modelled with the Witness simulation. The road network transport parameters are defined to conduct the dynamic adjustment of path parameters for different transport schemes. The visible interface of the Witness simulation illustrates the running state and relevant data of container goods at transport nodes and transport paths, and provides reference data for transport scheme selection and road network optimization. The multi-objective "Multimodal Transport Efficiency Evaluation" (MTEE) framework for measuring the efficiency of multimodal transport routes is developed based on the preferences of decision makers. The weight range for the four indexes of the transport scheme is determined by an order relation analysis method, i.e., the G1 method (Li et al., 2017). The final weight was determined by a combined G1 method and entropy-weight fuzzy analytical hierarchy process (E-FAHP) method. The method evaluates each transport scheme, and optimizes the multimodal transport line. This overcomes the arbitrariness and subjectivity in the selection of a multimodal transport scheme, and adds the evaluation index for energy conservation and emission reduction.

The rest of the paper is organized as follows. Section 2 establishes the Wuhan-Berlin road network structure as the research area, selects the transport efficiency evaluation index, and constructs the simulation MTRN model and the MTEE framework. In Section 3, the proposed approach is applied to the Wuhan-Berlin section of the China Rail (CR) Express. Section 4 and Section 5 provide the discussion and conclusion, respectively.

2. Data and methodology

This section includes the case background, evaluation indicators, MTRN model, and MTEE framework.

2.1. Case background

With the initiative of "One Belt One Road" and the development strategy between China and Europe, the trade volume is increasing significantly every year. In 2018, 23.498 million twenty-foot equivalent units (TEU) were transported through international shipping between China and Europe (Ministry of Transport of the People's Republic of China, 2019). More than 600,000 TEU were transported via land transport by the CR Express. The enormous volume of goods between China and Europe implies massive energy consumption and carbon emissions. The selection of the freight transport mode between China and Europe was mainly based on the economic benefits of each transport mode, rather their energy-saving and emission reduction benefits. Sea transport has remained the main mode of transport between China and Europe, owing to its complete supply chain and uniform trade rules, as well as the low transport cost. In terms of green transport, however, water transport is not considered as the best way to transport goods. Sea freight is suitable for low added-value goods close to the coast, whereas the China-Europe railway line is more suitable for high added-value goods in inland regions. This study establishes a "green" evaluation index for multimodal transport routes, and provides reasonable plans for modes and routes of transporting goods to promote sustainable transport development.

As a hub of logistics and goods in China, Wuhan has become an

important window for China-Europe trade. The CR Express (Wuhan), radiating across 28 countries and more than 60 cities, includes Germany, a manufacturing country, and Russia, a resourceful country, as its destinations. Wuhan, involving the unique multimodal transport network, has the fifth-largest transport volume among the CR Express cities. This study chooses Wuhan, China and Berlin, Germany as the origin and destination, respectively. Five cargo containers, totaling 100 tons, are assumed to be transported. In the transport section from Wuhan to Berlin, a total number of 12 transport nodes are involved (including 4 overseas nodes), and 34 transport subsections can be operated. There is no special requirement for passing by a node, and the transport network diagram shown in Fig. 1 is based on a one-way transport principle. The basic information of the nodes in the multimodal transport network is shown in Table 1.

2.2. Evaluation indicators

Energy conservation, emission reduction, and operation efficiency are selected the indexes for the efficiency evaluation of multimodal transport. The operation efficiency of multimodal transport refers to the multimodal transport time (MTT), multimodal transport cost (MTC) and other factors involved in the transport process. Energy conservation and emission reduction refer to the multimodal carbon emissions (MCE) and multimodal energy consumption (MEC), which can effectively reveal the comprehensive benefits of different transport routes in the multimodal transport network.

1) Transport time - the multimodal transport time is affected by multi-attributes, such as the transport distance, transport mode, traffic conditions, environmental attributes, transport policy and transfer time of goods in the transport process.
2) Transport cost - the multimodal transport cost is the cost incurred in the transport process. It mainly includes the vehicle cost between nodes, the transshipment cost on nodes and the time cost caused by waiting on nodes.
3) Energy conservation and emission reduction- Energy conservation and emission reduction refer to the carbon emissions and comprehensive energy consumption in the transport process. The comprehensive energy consumption must be converted to standard coal, as different energies would be consumed for each transport mode.

The carbon emissions can be estimated by Equation (1):

$$C_e = \sum_{i=1}^{n} \left(L_i * M * e_K^P * C_E * C_F \right) \quad (1)$$

L_i represents the transport distance for transport sections; M represents the total weight of a transportation; e_K^P represents a carbon emission coefficient for different transport modes; C_E represents the heating value, which is set as 8800 kcal/t; and C_F represents the emission coefficient, which is set as 0.00031 t/kcal in this study.

The container in this study is a 40-foot general-purpose container, and the basic information of each transport mode is shown in Table 2.

The basic information for transferring between different transport models is shown in Table 3.

Data source: the data source are from Unified Regulations on Transit Prices for International Railway Freight Intermodal Transport; Milestone of the World's Major Ports; China Railway Atlas; Rapid Mapping of China Highway Network; 2006 IPCC Guidelines for National Greenhouse Gas Inventory; Global Energy & CO2 Status Report; The CR Express Construction and Development Plan (2016–2020).

2.3. The simulation MTRN model

The MTRN simulation model was built in the Witness system based on parameters concerning the network, section, station and other factors. A traditional optimization algorithm would assign default values to the transportation parameters. In contrast, the MTRN model involves different assignments to the transport parameters, which can realize the path parameter transformation caused by transport scheme adjustment. It is able to illustrate the operation state and relevant data of container goods in the network. The simulation generates the transport time, distance, cost, carbon emissions, and comprehensive energy consumption for each transportation route.

To clearly describe the MTRN simulation model, the definitions

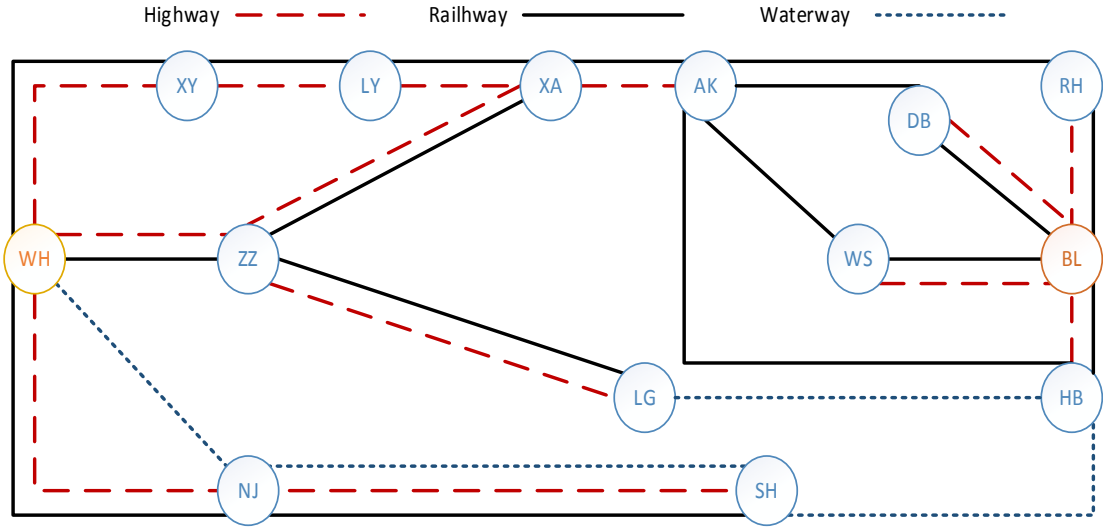


Fig. 1. The transport network structure in Wuhan-Berlin.
Notes: WH, Wuhan; XY, Xiangyang; LY, Luoyang; XA, Xian; NJ, Nanjing; ZZ, Zhengzhou; SH, Shanghai; LG, Lianyungang; AK, Alashankou; WS, Warsaw; RH, Rhodes; DB, Duisburg; HB, Hamburg; BL, Berlin.

Table 1
Basic information of the nodes in multimodal transport network.

Section & Distance/km		Section & Distance/km		Section & Distance/km	
WH—XY	314 (R)	WH—ZZ	659 (R)	WH—NJ	516 (R)
	351 (H)		724 (H)		527 (H)
XY—LY	385 (R)	ZZ—XA	511 (R)		696 (W)
	349 (H)		530 (H)	NJ—SH	294 (H)
LY—XA	387 (R)	ZZ—LG	584 (H)		301 (R)
	371 (H)		754 (R)		347 (W)
XA—AK	3045 (R)	LG—HB	20,370 (W)	SH—HB	18,848 (W)
	3271 (H)	AK—HB	6689 (R)	AK—RH	6361 (R)
DB—BL	859 (R)	WS—BL	780 (R)	AK—DB	6955 (R)
	735 (H)		1158 (H)	AK—WS	6278 (R)
RH—BL	460 (R)	RH—BL	1049 (H)	HB—BL	375 (R)
					289 (H)

Notes: R, Rail transport; H, Highway transport; W, Waterway transport.

Table 2
Basic information for different transport mode.

	Transport speed (km/h)	Transport cost	Multimodal carbon emissions (MCE) coefficient (kg/tkm)	Multimodal energy consumption (MEC) (tce/100 tkm)
R	35	R_1: 680 ¥/km R_2: 2.574 ¥ R_3: 1.122 ¥	0.468	0.396
H	47.51	H_1: 9 ¥/km H_2: 46 ¥	0.733	2
W	12.5 (domestic) 27.5 (foreign)	W_1: 0.63¥/km	0.592	0.41

Notes: h, hours; R_1, Railway base price 1; R_2, Railway base price 2; R_3, construction fund fee; H_1, Unite highway transport price; H_2, Packing charge; W_1, Unit waterway transport price; tce, ton of standard coal equivalent.

Table 3
Transfer coefficients between different transport mode.

	Transport costs (¥/t)	Transport time (h/TEU)	MCE coefficient (kg/TEU)
R–H	5.5	0.2	4.05
H–W	6	0.4	4.25
W—R	7.5	0.37	4.65

of symbols and variables are provided as follows.

Q—Number of containers
$P_{(X,Y)}$—Transport mode for the Yth section in the Xth transport scheme
T_X—Running time for the Xth transport scheme
T_{XB}—System entering time for container in the Xth transport scheme
T_{XS}—System leaving time for container in the Xth transport scheme
L—Transport distance for the Xth transport scheme
$L_{(X,Y)}$—Transport distance of the Yth section in the Xth transport scheme
C_X、E_X、N_X—Transport cost, carbon emissions, and comprehensive energy consumption in the Xth transport scheme
$C_{(X,Y)}$、$E_{(X,Y)}$、$N_{(X,Y)}$—Transport cost, carbon emissions, and comprehensive energy consumption for the Yth section in the Xth transport scheme
$CS_{(X,Y)}$、$ES_{(X,Y)}$—Transport cost and carbon emissions for the Yth transfer node in the Xth transport scheme
c_k^P、es_k^P、n_k^P—Transport cost, carbon emissions, and comprehensive energy consumption of container in transfer node
Loadtime—Transit transfer time of container in transit transfer node
P_q—Other costs for rail transportation, highway transportation, and waterway transportation
$P_{(X,Y)}$—Transport mode selection parameters of for the transit transfer node

The multi-objective function of the MTRN model is expressed as follows:

$$T_X = T_{xS} - T_{xB}\ ;\ \ L = \sum_{y=1}^{n} L_{(x,y)} \tag{2}$$

$$CS_{(X,Y)} = c_k^P * M + P_{q2} \tag{3}$$

$$ES_{(x,y)} = es_k^P * Q \tag{4}$$

$$N_x = \sum_{y=1}^{n} n_k^P * L_{(X,Y)} \tag{5}$$

$$P_4:\ P_{(X,Y)} = H(R);\ P_{(X,Y+1)} = R(H) \tag{6}$$

The transport time and distance are estimated by Equation (2). The transport cost, carbon emissions, and comprehensive energy consumption are estimated by Equations (3)–(5). The transport mode shift in function P is illustrated using Equation (6).

The basic framework of the MTRN simulation model is shown in Fig. 2.

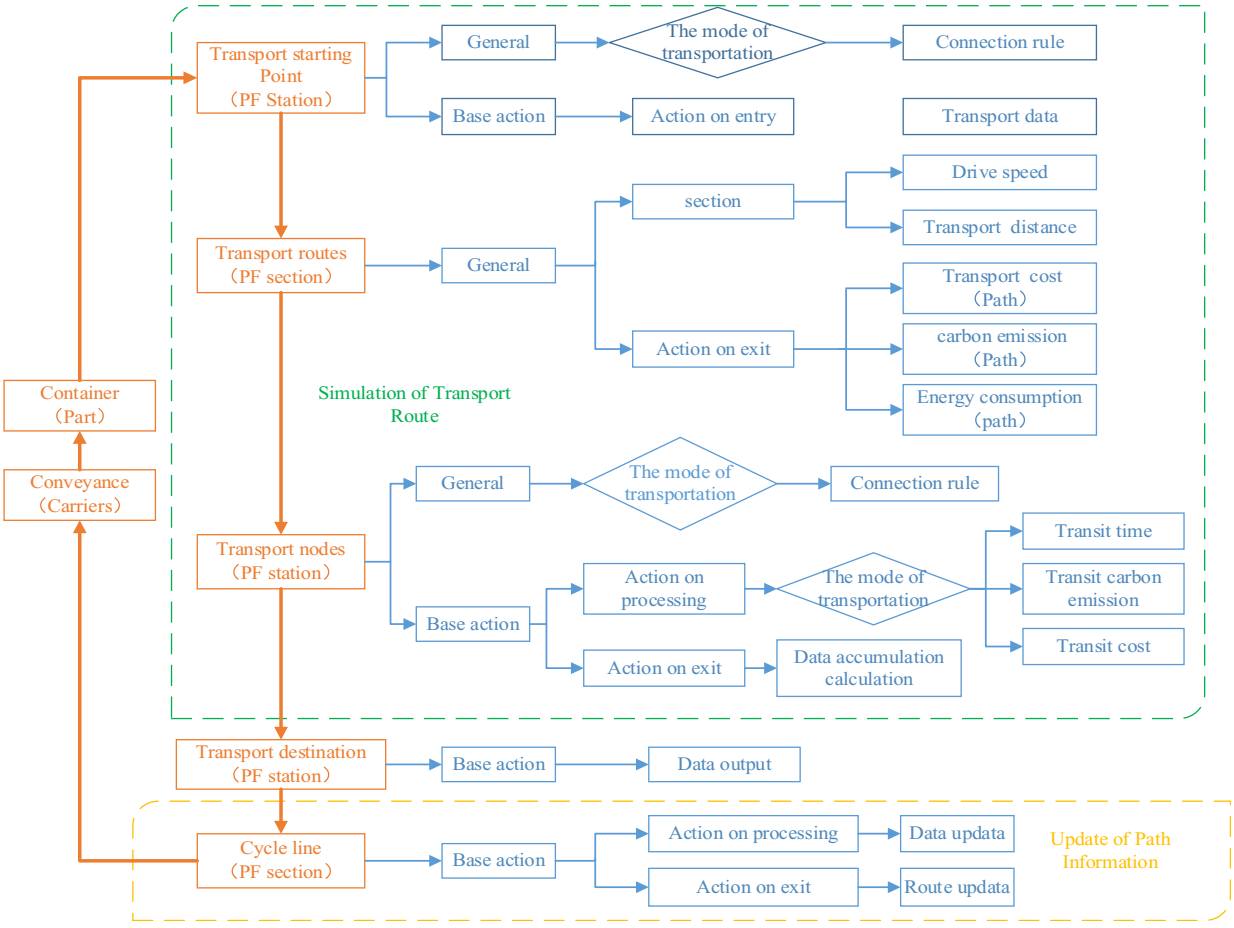


Fig. 2. The basic framework of the MTRN simulation model.

2.4. The MTEE framework

The efficiency evaluation for a multimodal transport route emphasizes the transport time, transport cost, carbon emissions, and comprehensive energy consumption. In contrast to a single-objective method, the variable weight and the index difference must be considered in the multi-objective decision-making process. There is a benefit-reverse relationship among indexes. The evaluation weight could be influenced by the preferences and requirements from multimodal transport decision-makers, such as managers, operators, and shippers in the transport process. This study aims to incorporate the transport mode performance, evaluation indexes' interactions, and decision makers' preferences in the decision-making process for conducting the efficiency evaluation and the route choice. The MTEE framework is established while considering different transport schemes and opinions from multimodal transport participants. The G1 method is used to generate the initial evaluation index weight according to fuzzy influencing attributes, and then the initial weight is adjusted through the E-FAHP method. The combination of the G1 method and E-FAHP method provides a more objective and accurate assessment. The evaluation structure is shown in Appendix A.

2.4.1. G1 method

The G1 method is an index preference method, and directly illustrates an opinion from a decision maker. The index weight range is more precise, and more transport schemes are considered.

1) Determine the Order Relation

The evaluation index set $X = \{x_1, x_2, \cdots, x_m\}$ is composed of M evaluation indexes, and the order relationship is determined according to the relative importance of the evaluation indexes in different transport schemes.

2) Calculation of the Index Importance

By comparing the relative importance between different indexes x_{j-1} and x_j, the rational judgment regarding the importance ratio ω_{k-1}/ω_k of the evaluation index x_{j-1} to x_j is determined as:

$$\omega_{k-1} / \omega_k = r_k \quad (k = m\ ,\ m-1\ ,\ m-2\ ,\ \cdots 3\ ,\ 2) \tag{7}$$

When m is large, $r_m = 1$, according to the order relation.

3) Subjective weight (ω_k) Determination

$$\begin{cases} \omega_k = \left(1 + \sum_{k=2}^{m} \prod_{i=k}^{m} r_i\right)^{-1} \\ \omega_{k-1} = r_k \omega_k \ \ (k = m\ ,\ m-1\ ,\ \cdots\ ,\ 3\ ,\ 2) \end{cases} \tag{8}$$

2.4.2. E-FAHP method

A fuzzy analytical hierarchy process is commonly utilized to determine the hierarchy in a multi-objective evaluation problem. This study assigned index scores to different transport schemes

according to the preferences of multimodal transport participants. The entropy weight is used to build the judgment matrix, which is adjusted by the fuzzy analytical hierarchy process method.

1) According to the layer diagram, the index score, ranging from 0.1 to 0.9, is normalized as in Equation (9).

$$\text{Negative index}: v_{ij} = max(r_j) - r_{ij} / max(r_j) - min(r_j) \tag{9}$$

2) Calculate the proportion of the index j of object i (p_{ij}), and the entropy value of index j (e_j).

$$\begin{cases} p_{ij} = v_{ij} \Big/ \sum_{i=1}^{m} v_{ij} \\ e_j = -1 \Big/ ln(m) \sum_{i=1}^{m} p_{ij} \cdot lnp_{ij} \end{cases} \tag{10}$$

3) Calculate the coefficient of variation(d_j), $d_j = 1 - e_j$ and the entropy weight(W_j).

$$W_j = d_j \Big/ \sum_{j=1}^{n} d_j \tag{11}$$

4) Adjust the fuzzy evaluation matrix in terms of the FAHP method.

$$\begin{cases} c_i = \sum_{j=1}^{n} f_{ij} \\ c_{ij} = \dfrac{c_i - c_j}{2n} + 0.5 \end{cases} \tag{12}$$

5) Determine the factor weights w_i for each level, according to the hierarchical ordering method.

$$\begin{cases} \overline{s_i} = \left(\prod_{j=1}^{n} r_{ij} \right)^{1/n} \\ w_i = \dfrac{\overline{s_i}}{\sum_{i=1}^{n} \overline{s_i}} \quad (i,j = 1,2,3,\cdots,n) \end{cases} \tag{13}$$

The combined weighting method is used to generate the final weight, which reduces the subjective impacts and provides practical evaluation results.

$$w_j = \alpha w_{j1} + \beta w_{j2} \tag{14}$$

w_j represents the combined weight; w_{j1} represents the G1 method weight; w_{j2} represents the E-FAHP weight; α and β respectively represent the weight pending coefficients, and were set as 0.5 in this study.

3. Results

3.1. Simulation results of the witness platform

This study applied the Witness platform for providing a dynamic simulation to build a multimodal transport network, and to manage the input and output data through a Microsoft Excel database. The road network from Wuhan to Berlin was built in the MTRN model, including 52 transportation logic elements, 14 transfer nodes and 34 transportation sections. The transport mode was defined by a group of dynamic variables, whereas the transport parameters in the network were determined by 27 system variables. The transport index was calculated using 9 dynamic variables in the multimodal transport route. 1090 variables were generated in Microsoft Excel database to obtain the transport parameter values for all routes.

The transport route set was divided into two categories: 192 land transport routes from China to Europe, and 26 traditional water transport routes. The transport parameters for the different routes were collected from real-world multimodal transport from China to Europe. From the simulation, the transport parameter distribution range for the available routes between Wuhan and Berlin is shown in Table 4.

The transport parameter frequency distribution for the available routes between Wuhan and Berlin is shown in Fig. 3.

3.2. Optimization for the multimodal transport route

Freight attributes and transport requirements are the major attributes influencing decision-makers' opinions in the transport process. 11 transport scenarios were established according to the decision-makers' opinions on the transport time, transport cost, energy conservation, and emissions reduction. The optimal routes under the various transport schemes generated from the MTEE are shown in Tables 5–7.

1) When a single objective is considered, the optimal multimodal routes from Wuhan to Berlin are shown in Table 5.

G1 method: The importance order for the efficiency index of multimodal transport is $x_{c1} \gg x_{c2} x_{c4} = x_{c3}$, and $r_2 = 1.6, r_3 = 1.7$, and $r_4 = 1$. The initial weight is computed as W_{j1}=(0.423,0.265,0.156,0.156).

E-FAHP method: Based on the weight of entropy W_E = (0.381, 0.492, 0.124) and the scale factors B_1, C_1, and C_2 of the FAHP index layer, the weight of the FAHP is calculated as W_{j2}=(0.445,0.2222,0.158,0.175).

The combined weight W= (0.434, 0.244, 0.157, 0.166). Through normalizing the transport index, the optimal route for the multimodal transport is generated. If the decision-makers prefer another index, the weight calculation approach is similar.

3) When multiple objectives are considered and the decision makers prefer two objectives, two-factor optimizations for the multimodal transport route can be generated, as shown in Schemes 8–10.

3.3. Comprehensive analysis

3.3.1. Sensitivity analysis

A sensitivity analysis was conducted for the 11 multimodal transport schemes.

(1) Different transport decision makers may have different opinions regarding transport time, transport cost, energy conservation, and emissions reduction, and the evaluation preferences and requirements in the transport process may also be different.

- More attention is paid to the development of low-carbon and green transportation when the decision makers are governors. The green transportation indicators, carbon emissions, and comprehensive energy consumption of transportation activities are generally desired for the evaluation of transport efficiency. The energy

Table 4
Parameter distribution range for available routes.

	MTT (h)	MTC (¥)	MCE (t)	MEC (tce)
Land transport	[300, 365]	[168,000, 293,000]	[7.42, 20.93]	[5.51, 10.38]
Water transport	[734, 785]	[85,049, 122,000]	[18.75, 21.87]	[8.20, 12.46].

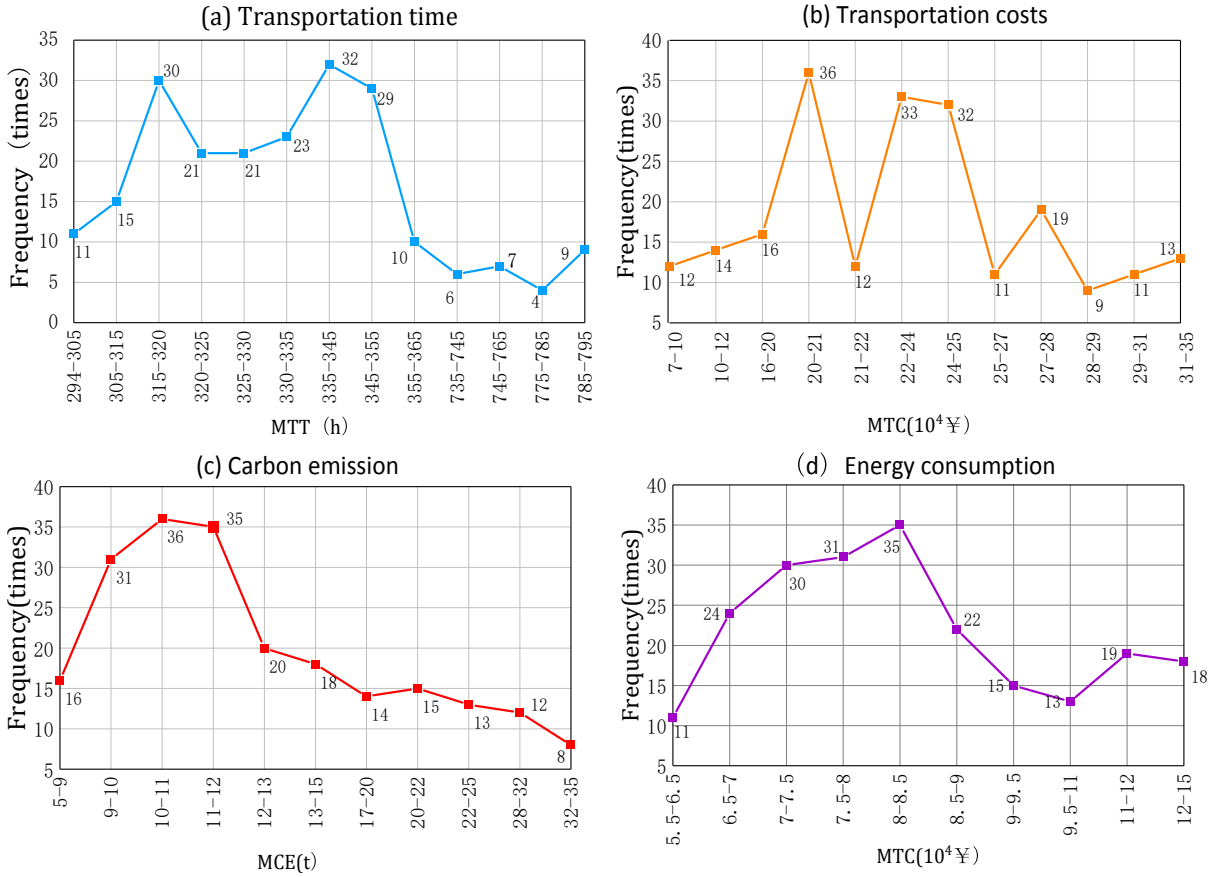


Fig. 3. Parameter frequency distribution for available transport routes.

Table 5
Optimal route for multimodal transport.
2) When multiple objectives are considered and the decision-makers prefer one objective, single factor optimizations can be generated for the multimodal transport route as shown in Schemes 5—7. When the decision makers have a strong preference regarding the transport time, the evaluation weight would be calculated as in Section 2.4 above.

Evaluation Index	Result	Optimal Route and Transport Mode
Scheme 1: optional MTT	294.44 h	WH—ZZ—XA—AK—RH—BL (H—H—H—R—H)
Scheme 2: optional MTC	71,482 ¥	WH—NJ—SH—HB—BL (H—H—W—W—H)
Scheme 3: optional MCE	7.93 t	WH—ZZ—XA—AK—RH—BL (R—R—R—R—R)
Scheme 4: optional MEC	5.508 tce	WH—XY—LY—XY—AK—DB—BL (R—R—R—R—R—H)

conservation and emission reduction indexes are the priority, as compared to the transport cost and time.

- The economic benefits in the transport process are the emphasis if the decision makers are multimodal transport operators. To obtain more benefit, the transport cost and time are the first and second priority in the optimization process of the transport route. The energy conservation and emissions reduction are paid slight attention.
- When the decision maker is a multimodal transport shipper, the focus is on fulfilling the transport task with the lowest cost within the limited time. The transport cost and time are still the first and second priority in the optimization process of the multimodal transport route. The energy conservation and emissions reduction are not considered.

Multimodal operators and shippers have similar preferences regarding the efficiency evaluation for multimodal transport. The transport cost and transport time are always more important than the energy conservation and emissions reduction, and energy consumption is the priority when compared to carbon emissions.

Table 6
Optimal route for multimodal transport through single factor optimization.

3) When multiple objectives are considered, the global optimization for the multimodal transport route can be generated as shown in Table 7.

Preference index & Optimal route	Evaluation Index	W_{j1}	W_{j2}	W	Simulation Result
Scheme 5:MTT	MTT	0.423	0.445	0.434	307.48 h
WH—XY—LY—XA—AK—HB—BL	MTC	0.265	0.222	0.244	207,319 ¥
H—H—H—R—R—H	MCE	0.156	0.158	0.157	10.454 t
	MEC	0.156	0.175	0.166	8.279 tce
Scheme 6:MTC	MTT	0.236	0.227	0.232	322.26 h
WH—ZZ—XA—AK—WS—BL	MTC	0.453	0.356	0.405	170,227 ¥
R—R—R—R—R	MCE	0.142	0.198	0.170	10.105 t
	MEC	0.169	0.219	0.194	8.508 tce
Scheme 7:MER	MTT	0.233	0.232	0.233	321.27 h
WH—XY—LY—XA—AK—WS—BL	MTC	0.240	0.256	0.248	201,851 ¥
R—R—R—R—R—R	MCE	0.256	0.237	0.247	8.816 t
	MEC	0.271	0.275	0.273	6.495 tce
Scheme 8:MTT&MTC	MTT	0.347	0.345	0.346	318.63 h
WH—ZZ—XA—AK—HB—BL	MTC	0.376	0.377	0.376	177,531 ¥
R—R—R—R—H	MCE	0.146	0.14	0.143	8.943 t
	MEC	0.131	0.138	0.135	6.880 tce
Scheme 9:MTT&MER	MTT	0.356	0.35	0.353	314.60 h
WH—XY—LY—XY—AK—WS—BL	MTC	0.177	0.187	0.182	204,992 ¥
H—H—R—R—R—R	MCE	0.213	0.208	0.211	9.571 t
	MEC	0.254	0.255	0.254	7.433 tce
Scheme 10:MTC&MER	MTT	0.155	0.202	0.178	349.80 h
WH—ZZ—XA—AK—RH—BL	MTC	0.366	0.403	0.385	168,963 ¥
R—R—R—R—R	MCE	0.262	0.188	0.225	7.936 t
	MEC	0.217	0.207	0.212	6.313 tce

Table 7
Optimal route for multimodal transport through global optimization.

Scheme 11:MTT&MTC&MER	MTT	MTC	MCE	MEC
WH—ZZ—XA—AK—WS—BL R—R—R—R—R—R	322.09 h	172,227 ¥	8.105 t	6.508 tce
WH—XY—LY—XA—AK—HB—BL H—H—R—R—R—G	311.73 h	195,538 ¥	11.042 t	7.766 tce
WH—ZZ—XA—AK—HB—BL R—R—R—R—H	318.63 h	177,531 ¥	8.943 t	6.880 tce

This is mainly caused by the fierce competition in the transport market, and the lack of incentives for low-carbon transport. Multimodal operators are more willing to promote their competitiveness without paying extra cost to improve the energy conservation and emissions reduction. This study is designed to use energy saving and emission reduction values as a compulsory index for multimodal transport route selection. The government can formulate corresponding policies to guide the transport market to promote beneficial environmental schemes.

(2) The fluctuation diagram of the transport parameters for various transport schemes is shown in Fig. 4.

Different combinations of transport routes and modes promote diversity in route selection. There are significant differences in the transport index among different transport schemes. In this study, several transport schemes, with similar transport times and transport cost performances, may provide extremely different energy conservation and emission reduction indexes. The land transport scheme WH-ZZ-XA-AK—HS—BL with the H-R-R-R-H mode and the scheme WH-XY-LY-XA-AK-HB-BL with the R-R-R-R-R-R mode provide similar transport cost and transport time, but the carbon emissions and comprehensive energy consumption decrease by 38% and 35% respectively, when the latter scheme is used. The scheme WH-ZZ-LG—SH—HB-BL with the mode R-R-R-W-R would increase the transport time and transport costs by 2.59 h and 6207 ¥ as compared to the traditional water transport scheme WH-ZZ-LG-HB-BL with the mode R-R-W-H. However, the carbon emissions can be reduced by 10.3%, and the comprehensive energy consumption can be reduced by 16.2%. The benefits of energy saving and emissions reduction for the China-European land transport lines are significant when the economic benefits are similar as compared to traditional water transport. An increase of transport nodes results in more flexible route selection, which enlarges the differences of various transport indicators.

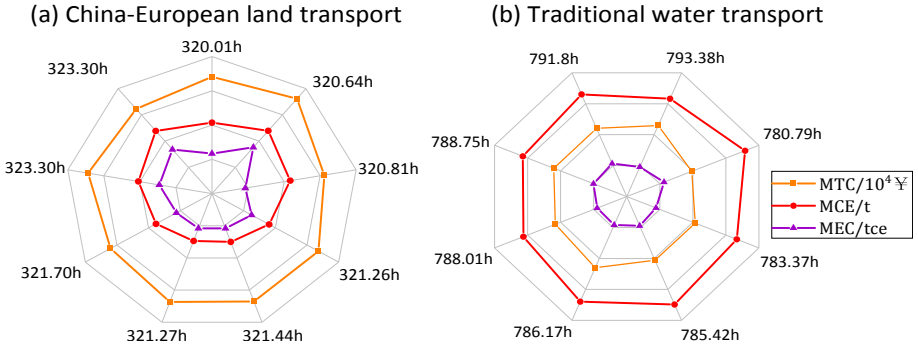


Fig. 4. Fluctuation diagram of various transport schemes.

(3) Reasonable multimodal transport planning simultaneously improves economic benefits, energy saving, and emissions reduction. However, various transport demands and preferences make it difficult to obtain an optimal scenario while considering only economic objectives. The simulation results of MTRN model provide the differences between transport indexes, demonstrating the diversity of routes. The efficiency evaluation method of the MTEE model is believed to optimize the multi-objective transport schemes in different transport scenarios, and to effectively promote the development of green transport. The differences in energy efficiency and emissions reduction for the optimal routes in different transport scenarios are shown in Fig. 5. The values of the energy saving and emissions reduction indexes can be reduced by up to 24% in the case that the economic benefit indicator meets the transport demand.

3.3.2. *Performance evaluation of the transportation route*

Table 8 illustrates a variation of transport parameters while comparing the transport route and transport mode.

The long-range transport between Wuhan and Berlin is mainly undertaken by rail and water transport; the highway and inland river transport play a connecting role. Besides Wuhan, other origin cities for the CR Express from China to Europe, i.e. Zhengzhou and Xi'an, can also be selected by the shipper according to the number of trains running, freight rate, and handling efficiency. Highway transport presents the advantage of being able to connect to any CR Express in short- and medium-range transport. As for long-range transport (over 300 km), highway transport does not have an advantage when compared to the CR Express.

In the overseas transport process, water transport, a green transport mode, has evident advantages in cost and carbon emissions. There are 26 water transport routes in the Wuhan-Berlin transport section. The water transport cost, which is approximately 50% lower than that of the CR Express, is also lower than 88% of the 218 simulated transport schemes. However, water transport is time-consuming, and the total transport time mainly ranged from 734 h to 785 h, would bear a 57% extra time loss. It is unstable and has a high rate of damage from being affected by natural conditions. Railway transport presents a massive advantage in long-range transport. The optimal route of overseas transport is AK-WS-HB-BL with the railway mode. The range of the transport time for optimal route is from 310 h to 330 h, whereas the transport cost is concentrated in the interval [182,637 ¥, 242,903 ¥], and the energy conservation and emissions reduction indexes are better than 77% of the other transport schemes.

The optimal land transport routes in the Wuhan-Berlin section are Route 2 and Route 3 in Table 8. These routes, with a higher transport cost, are suitable for high-value urgent freight. For traditional water transport in the Wuhan-Berlin section, Route 5 in Table 8 is recommended. This route is suitable for low-value and non-emergency freight.

This study uses a comparison to traditional water transport. The CR Express presents a disadvantage in terms of transport cost, but it could save over 60% of transport time, reduce carbon emissions by 40%, and realize directional transport and door-to-door service. This study is meaningful for strengthening land transport and promoting trade development between China and Europe.

4. Discussions

This section is included to build a green transport index for route selection, establish a new operation mode for the CR Express, and examine the applicability of the simulation model.

4.1. *Build the green transport index for route selection*

Taking the energy saving and emissions reduction as the evaluation indexes to select the multimodal transport scheme can reap

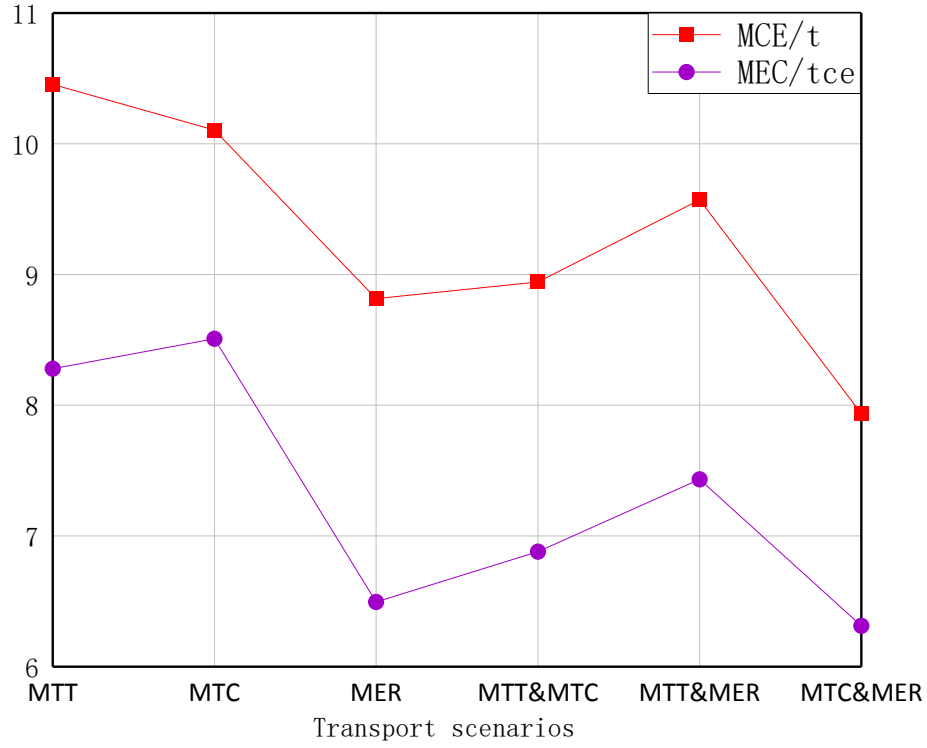
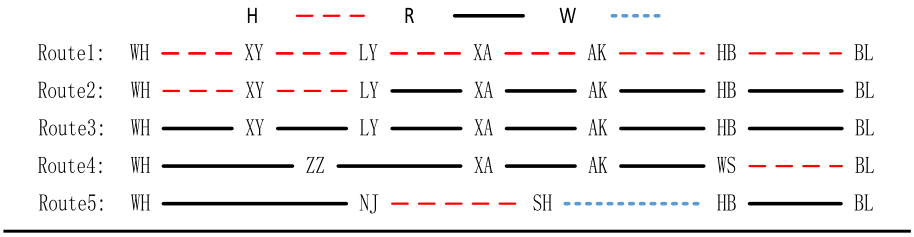


Fig. 5. The benefits in energy conservation and emissions reduction.

Table 8
The variation of transport Parameters.

Transport Route		MTT (h)	MTC (10^4¥)	MCE (t)	MEC (tce)
Domestic route	Route1	297.9	30.89	25.46	14.88
	Route2	314.59 (+16.69)	20.50 (-10.39)	9.57 (-15.89)	7.43 (-7.45)
	Route3	311.94 (+14.04)	20.66 (-10.23)	9.95 (-15.51)	7.91 (-6.97)
Foreign route	Route4	322.08	17.02	10.11	8.51
	Route5	720.03 (+397.95)	9.28 (-7.74)	19.28 (+9.17)	12.41 (+3.9)



better environmental benefits and promote the development of a green transport industry. In the past, the selection of multimodal transport schemes often paid attention to attributes such as cargo connection efficiency, transport time and cost, cargo attributes, government subsidies, and different transport requirements, whereas the environment-related energy saving and emission reduction targets were ignored. Through the analysis of the superior transport routes under different transport situations, an optimal transport scheme can not only improve the operation efficiency, but can also have effects on the comprehensive energy consumption and carbon emissions. This study builds a green transport evaluation index to select the optimal transport scheme, especially for long-range multimodal transport. That is mainly because the reduction in the comprehensive energy consumption and carbon emissions accumulate to a significant amount in long-distance transport.

Through the sensitivity analysis of various transport indexes, it is seen that multimodal transport operators and shippers focus on the transport time and cost when choosing an optimal transport scheme. As performers in the transport process, transport operators and shippers would gain economic benefits by reducing the transport time and cost, at the expense of the energy conservation and emission reduction indexes. To address this dire situation, the advocates of low-carbon and green transport, i.e. the governors in multimodal transport, should act to formulate policies and regulations to guide the development of multimodal transport in a green direction. The CR Express Construction and Development Plan (2016–2020) was issued two years ahead of the schedule, and it is recommended that multimodal transport governors pay more attention to the energy conservation and emissions reduction targets. Unlike prior subsidies utilized to solve a supply problem, the current subsidy should promote green multimodal transport by supporting transport routes and transport schemes with higher energy conservation and emissions reduction indexes. The transport subsidy policy should also be issued to guide the development of low-carbon transport and help manage the greenhouse effect.

4.2. Establish a new operation mode for the CR express

There are competition and overlap among the transport routes of the CR Express in China, which reduces the transport efficiency and competitiveness as compared to the water transport. In 2018, the average number of containers transported by CR Express was 96 TEU, whereas the double-deck container trains in the United States have 150 carriages per train and can transport 300 53-foot containers, approximately 800 TEU. The transport capacity of container trains in the CR Express is only 1/8 of that for the U.S. trains, and is caused by the repeated lines and competition. According to statistics, there are 59 cities and 65 transport routes in the CR Express (Wang et al., 2018a, b; Jiang et al., 2018), resulting in the insufficient supply of freight and a low number of transport containers for a train.

The scattered distribution of the transport routes makes it difficult to improve the transport efficiency and provide non-stop transport service. Therefore, the results suggest reforming the operation mode for the CR Express through reorganizing the transport routes and freight resources. The large container distribution centers for the CR Express between China and Europe should be built adjacent to the national boundaries, and the trains from different domestic cities should be reorganized and rescheduled to provide high capacity and non-stop transport service. The return trains should also be reorganized and rescheduled to make full use of the transport capacity. The implementation of this operation mode can eliminate the competition among the domestic lines of the CR Express, and solve the freight source shortage problem and transport capacity waste problem for individual trains. The reorganized trains can also realize the direct delivery of freight. The proposed operation mode plays a positive role in promoting the competitiveness of CR Express, while reducing transport cost and carbon emissions.

4.3. Applicability of the simulation model

Multimodal transport is a complicated process involving various transport elements. The advantages of using Witness software to construct the road network structure is that it accurately defines different parameters for dynamic simulation of a transport process, and improves the accuracy of simulation results. The dynamic visibility view in Witness software clearly expresses the transport status of freight in the system. The MTEE model is designed by combining the G1 and E-FAHP methods, in which different transport scenarios are constructed, and the influence of subjective judgment is reduced. The final optimal transport scheme is more applicable. This model only considers the basic parameters of time, cost, carbon emissions, and comprehensive energy consumption in the transport process. However, a more complicated design is

possible. By adding new variables to the model such as road capacity, transportation risk, and service level, or further refining the transport process, the simulation results can be made more similar to an actual transport process. The multimodal transport efficiency evaluation model based on Witness software in this study is also applicable to transport planning problems such as supply chain modelling and route design.

5. Concluding remarks

Through the dynamic simulation of transport parameters in Witness software, this study proposed a MTEE framework for managing the route choice problem in multimodal transport. In the decision-making process, a combined G1 method and E-FAHP method was utilized to evaluate the efficiency of multimodal transport routes. The Wuhan-Berlin multimodal transport section was selected as a case study, and the proposed approach illustrated the process for constructing various transport schemes, which were optimized through the MTEE framework. The major findings are concluded as follows.

- There are attributes influencing the choice of the multimodal transport route, and a single transport scenario has difficulty meeting the development needs of green transport. When the transport scenarios are constructed according to different preferences and demands of intermodal transportation participants, the optimal routes are also different. A multi-objective and multi-scenario route selection method is believed to effectively promote the development of green multimodal transport.
- Transportation participants have different sensitivities to the evaluation indexes for transport time, transport cost, energy conservation and emissions reduction. Some decision makers prefer the operation efficiency to the energy conservation and emissions reduction indexes, and emphasize the transport time and transport cost while ignoring the energy conservation and emissions reduction benefits. However, the optimal transport scheme can reduce 35% of carbon emissions and save 38% of the comprehensive energy consumption when compared with the worst transport scheme. The carbon emissions difference between transport schemes can also be increased with a longer transport distance.
- Railway transport has advantages in energy saving and emission reduction in the Wuhan-Berlin road network, but the high transport cost is a disadvantage. The transport cost for the CR Express increases one time over the costs of water transport, yet the transport time is saved by 60%, and the carbon emissions can be reduced by 40%. This study suggests reforming the operation mode through reorganizing the transport routes, and establishing large container distribution centers to reduce operating costs. This study promotes a freight transportation transformation between China and Europe to a low-carbon and energy-saving mode.
- The parameters utilized in the simulation of the Wuhan-Berlin transport section were simplified according to the real world. The influence of uncontrollable attributes, such as environmental and policy impact on the transport capacity and transport efficiency, have been ignored, and could be investigated in further studies.

Author contribution

We would like to submit the enclosed manuscript entitled "Modelling the Route Choice Efficiency of Multimodal Transport with Witness Simulation".

Qing-Zhou Wang is guiding this study to be completed.
Jia-Meng Chen draft the first version.
Ming-Lang Tseng finalized the manuscript.
Hai-Min Luan draft the first version.
Mohd Helmi Ali helps to finalize this manuscript.

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Appendix A. The Evaluation Structure

Target Layer (A)	Factor Layer(B)	Indicator Layer(C)
Multimodal transport efficiency evaluation (A1)	Operational efficiency (B1)	Transport time (C1)
		Transport costs (C2)
	Energy conservation and Emission reduction (B2)	Carbon emission (C3)
		Comprehensive energy consumption (C4)

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